

Article

Optimizing Evaluation Systems for Industrial Land Inefficiency: A Pattern-Sensitive Framework Integrating Expert Knowledge and Machine Learning

Wei Cai ^{1,2} , Xin Zhang ¹, Fengjue Huang ³  and Mingyu Zhang ^{1,4,*}

¹ Department of Regional and Urban Planning, Zhejiang University, 388 Yuhangtang Road, Hangzhou 310058, China; caiwei2004@zju.edu.cn (W.C.); 3220101171@zju.edu.cn (X.Z.)

² Center for Balance Architecture, Zhejiang University, 148 Tianmushan Road, Hangzhou 310028, China

³ School of Urban Planning and Design, Shenzhen Graduate School, Peking University, 3688 Nanhai Avenue, Shenzhen 518067, China; huangfengjue@stu.pku.edu.cn

⁴ The Architectural Design & Research Institute of Zhejiang University Co., Ltd., 148 Tianmushan Road, Hangzhou 310028, China

* Correspondence: zhangmy18@zju.edu.cn

Abstract

The evaluation of inefficient industrial land is crucial for sustainable urban renewal, yet conventional methods are often compromised by applying a single uniform set of evaluation criteria that ignore local contextual patterns. We introduce a novel, pattern-sensitive framework that identifies distinct inefficiency patterns by interrelationships between evaluation indicators and land performance and calibrates expert-derived weights with data-driven insights. Using public access data for Xiaoshan District, Hangzhou, we establish an evaluation system via the Analytic Hierarchy Process (AHP). Subsequently, a novel iterative clustering method partitions parcels into segments sharing the same inefficiency pattern. Within each segment, a random forest model learns the local interrelationships from the data. This machine-learned information is then used to optimize the initial AHP weights, creating a unique evaluation system for each identified pattern. Results demonstrate that our optimization framework achieves Pearson correlations of 0.66–0.82 with ground-truth inefficiency across four identified patterns, outperforming traditional AHP-based models. Temporal validation on 2023 data confirms robustness of weights optimized on 2022 data, maintaining significant positive correlations (Pearson's $r = 0.58$ – 0.66) with ground-truth inefficiency across all segments. By synergizing expert knowledge with machine learning, this study provides an accurate tool to formulate targeted urban renewal strategies that move beyond one-size-fits-all solutions.

Keywords: industrial land inefficiency; pattern-sensitive evaluation system; weight optimization; urban renewal



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1. Introduction

Over the past four decades, China has undergone a rapid urbanization that has significantly altered its socioeconomic structure [1]. The nation's urbanization rate reached 66.16% by the end of 2023, a process largely driven by the extensive conversion of rural land into urban areas to fuel economic expansion [2,3]. While this approach drove substantial economic growth, it has now encountered significant physical and ecological constraints, leading to challenges such as land resource depletion, environmental degradation, and inefficient urban spatial structures [4,5]. In response, China's urban development strategy

has pivoted from this incremental expansion towards a new paradigm focused on the regeneration of existing urban land, a policy known as “stock development” or “urban renewal” [6,7]. This policy shift positions the redevelopment of underutilized and inefficient land from a secondary concern to key priority for achieving high-quality, sustainable urbanism [8,9].

Within the context of urban renewal, industrial land constitutes a particularly critical and complex issue [10,11]. As a legacy of the previous era’s extensive growth model, vast tracts of industrial land often occupy valuable urban space yet are characterized by low output, outdated functions, and significant negative externalities [12–14]. Consequently, the precise identification, evaluation, and redevelopment of these inefficient industrial parcels have become a focal point for implementing urban renewal projects [15]. The case of Xiaoshan District in Hangzhou, a major manufacturing hub navigating the dual pressures of industrial upgrading and urban expansion, serves as a representative case for this challenge, making it a suitable subject for the study.

A primary obstacle in addressing industrial land inefficiency has been the deficiencies of traditional data sources. Research on urban land use efficiency has historically relied on official statistical yearbooks, which typically aggregate data at spatially coarse administrative scales such as the provincial or municipal level [16,17]. This aggregation obscures the substantial intra-urban heterogeneity in land performance, precluding granular, parcel-level analysis. The recent proliferation of open-source geospatial big data, however, offers a promising opportunity to overcome this limitation. Data streams such as OpenStreetMap (OSM) for road networks, Points of Interest (POIs) for land use functions, and satellite-derived Nighttime Light imagery for economic activity now facilitate scalable and replicable analyses at the micro-scale of individual land parcels [15,18].

Despite these data advancements, methodological challenges persist in the accurate evaluation of land use performance. One dominant approach relies on expert-driven frameworks like the Analytic Hierarchy Process (AHP), which systematically incorporates qualitative domain knowledge but is hampered by inherent subjectivity and, more critically, overlooks spatial heterogeneity by employing a uniform set of weights across an entire study area [19,20]. This “one-size-fits-all” assumption is inconsistent with the complex spatial dynamics of urban systems [21]. At the other end of the spectrum are objective, data-driven methods like Data Envelopment Analysis (DEA) and the Entropy Weight Method (EWM). While these approaches reduce subjectivity, their weighting mechanisms are often formulaic and can fail to adequately capture the underlying data structure, potentially yielding results that are misaligned with real-world contexts and policy goals [22,23]. These limitations highlight a clear research gap: the need for a hybrid, pattern-sensitive framework that integrates the contextual insights of expert knowledge with the objective, localized patterns derived from data.

To address these challenges, this study introduces and validates a novel framework for constructing a set of evaluation systems to identify inefficient industrial land at the parcel scale, which is the primary contribution of this paper. Central to this framework is a data-driven algorithm designed not to replace expert knowledge but to calibrate it for local contexts. The methodology commences with an initial AHP-based evaluation system. It then partitions parcels into distinct segments using an iterative linear-fit clustering algorithm that groups them based on shared land performance response patterns, thereby identifying zones with distinct inefficiency patterns. Within each segment, a Random Forest model is employed to learn objective relationships from local data, and these insights are subsequently utilized to optimize the initial expert weights. This synthesis of AHP and machine learning generates a set of unique, pattern-sensitive evaluation systems, each tailored to a specific inefficiency pattern.

2. Literature Review

The evaluation of industrial land use efficiency is fundamental to sustainable urban management. Research on this topic has developed along two primary streams: macro-scale econometric models that assess efficiency at the administrative level, and micro-scale multi-criteria decision analysis (MCDA) that evaluates performance at the parcel level, typically through the construction of evaluation systems.

The first stream of research primarily employs econometric models to measure land use efficiency across large-scale administrative units such as cities or provinces. This stream includes non-parametric methods such as Data Envelopment Analysis (DEA) and its variants (e.g., Slack-Based Measure), and parametric methods such as Stochastic Frontier Analysis (SFA) [24–27]. These models are commonly employed to assess the relative efficiency of decision-making units by comparing their inputs (e.g., land area, capital) to their outputs (e.g., GDP, industrial output) [1,28]. For instance, Yi and Ma [29] applied DEA to assess the land use efficiency of Chinese provincial capitals, while Li et al. [30] utilized it to reveal significant spatial clustering of efficiency in cities along the Yellow River Basin. Although recognized for their objectivity in comparing production frontiers, these approaches operate at an aggregated spatial scale. This aggregation obscures the substantial intra-urban heterogeneity in land performance, rendering the findings unsuitable for guiding the granular, parcel-specific interventions required for effective urban renewal [16].

To address the scale limitations of macro-level models, a second stream of research employs micro-scale Multi-Criteria Decision-Making (MCDM) methods, which have become more feasible with the advent of geospatial big data. The availability of high-resolution, open-source data such as OpenStreetMap (OSM) road networks and Points of Interest (POI) has enabled researchers to move beyond coarse administrative boundaries to analyze individual land parcels [15,18,31,32]. Within this stream, the Analytic Hierarchy Process (AHP) is a dominant method due to its ability to structure complex problems and integrate diverse, often qualitative, criteria through expert judgment [19,33]. However, its reliance on subjective pairwise comparisons is a well-documented source of potential bias [22]. To address this, numerous hybrid models have been developed. A common approach is to combine AHP with objective weighting techniques like the Entropy Weight Method (EWM), which assigns weights based on the variance of each indicator [33], or with other MCDM techniques like TOPSIS [20,23]. Other studies have used Hierarchical Bayesian Networks (HBN) to refine expert-derived weights [34].

Despite these refinements, two fundamental research gaps persist. First, the objective adjustments made by methods such as the Entropy Weight Method (EWM) are often formulaic and context-blind. The EWM, for instance, determines an indicator's weight based on its statistical dispersion across all samples, not on its actual predictive relationship with the evaluation target (i.e., land inefficiency). This means an indicator with high variance will be assigned a high weight, regardless of whether that variance is meaningfully correlated with performance. Such approaches fail to learn from the underlying data, producing weights that may be disconnected from the real-world drivers of inefficiency [22,23]. Second, and more critically, AHP and its conventional hybrid variants produce a single, globally applied set of indicator weights. This “one-size-fits-all” approach operates on the flawed assumption that the determinants of inefficiency are uniform across all parcels. In reality, urban systems contain distinct archetypes of inefficiency, each driven by a different combination of factors, by applying a single evaluative lens, conventional methods obscure these crucial differences, leading to inaccurate assessments and one-dimensional policy recommendations [21].

To address these gaps, this study makes three main contributions. First, it proposes a replicable, open-source framework for parcel-level land performance evaluation that lever-

ages publicly available geospatial big data, providing the granularity needed for targeted urban renewal. Second, it introduces a novel segmentation method which partitions parcels based on latent interrelationships between indicators and land use inefficiency, rather than geography locations, to identify land use inefficiency patterns. Third, it introduces a hybrid weight optimization method that synergizes expert knowledge with machine learning. For each identified pattern, the framework learns objective relationships from the local data distribution to calibrate the initial, subjective expert weights, thus producing a more robust and empirically grounded evaluation system that integrates both domain expertise and data-driven evidence. The framework of this study is illustrated in Figure 1.

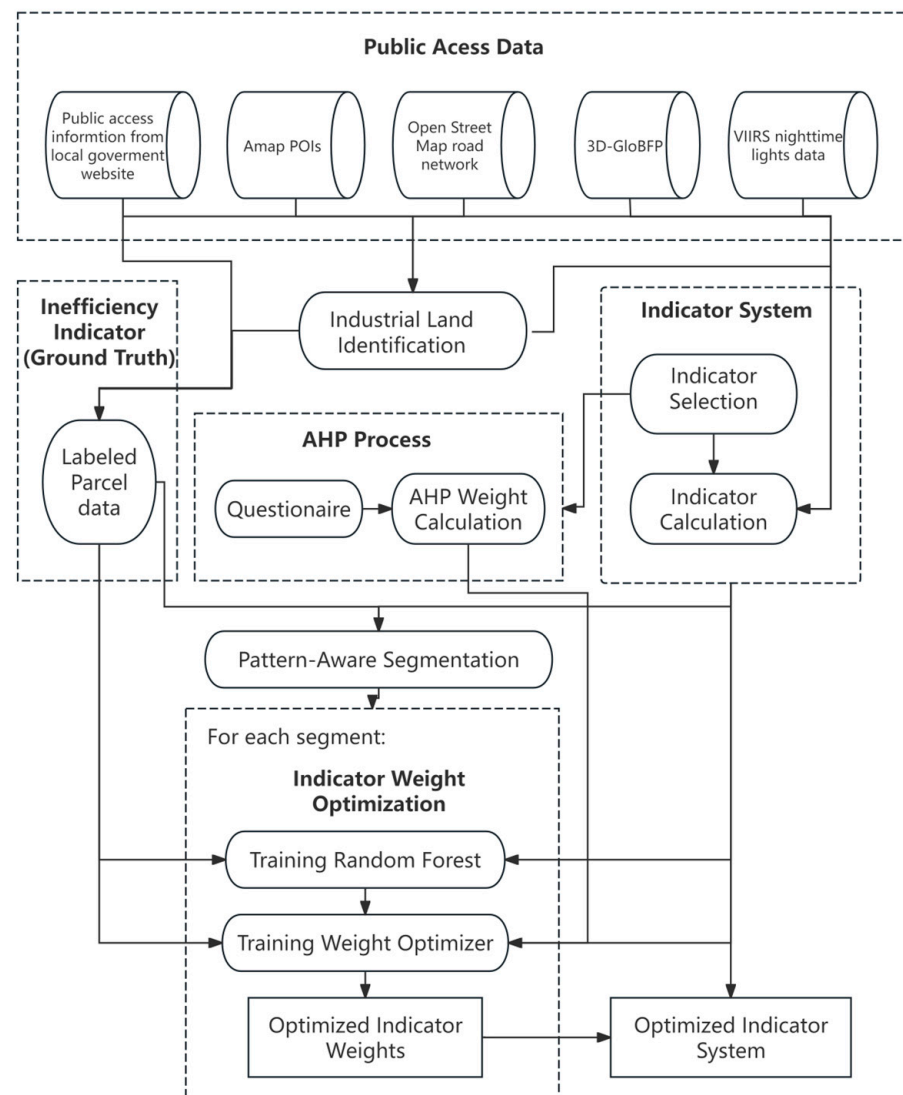


Figure 1. The research framework.

3. Materials and Methods

3.1. Research Area

Xiaoshan District in Hangzhou was selected as the test field for validating our methodological framework owing to its representative industrial landscape and prominent status as a key manufacturing hub in the Yangtze River Delta. In 2022, its industrial output surpassed 500 billion RMB, constituting over one-third of Hangzhou's total and underscoring the critical importance of its land use efficiency. The geographical location of the research area is illustrated in Figure 2.

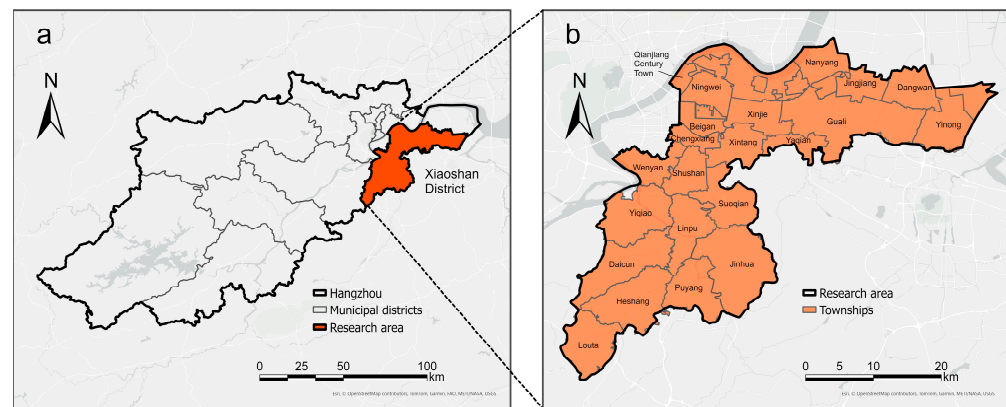


Figure 2. The research area, (a) the spatial location of the research area in Hangzhou, (b) the township distribution in the research area.

The industrial land in Xiaoshan exhibits pronounced spatial differentiation. In the northwest, areas adjacent to Hangzhou’s urban core, such as Ningwei townships, are undergoing a transition towards high-end manufacturing and R & D. Townships in the central-western and central regions, such as Xinjie and Guali respectively, are predominantly characterized by traditional manufacturing and contain notable concentrations of inefficient land parcels. The southern townships, including Linpu and Yiqiao, feature older industrial clusters from past extensive development. The district’s pronounced spatial differentiation, encompassing areas transitioning to R & D, traditional manufacturing zones, and legacy industrial clusters, provides the adequate heterogeneity necessary to rigorously test our framework designed to detect diverse land use performance patterns.

3.2. Industrial Land Identification and Land Use Inefficiency Indicator

This study utilized the OpenStreetMap (OSM) road network to delineate land parcels in Xiaoshan District. The process involved acquiring the 2022 OSM road data, which, after cleaning and manual refinement, provided the basis for partitioning the study area. The resultant polygons were subsequently filtered by area ($>30,000 \text{ m}^2$) to eliminate highly fragmented parcels and manually inspected to remove unsuitable geometries. Following delineation, industrial parcels were identified based on an analysis of Points of Interest (POIs) data. For each parcel, we calculated the Frequency Density (FD) and Category Ratio (CR) for the various POI types located within its boundaries. The metrics are defined as follows:

$$FD_i = N_i / S_i \quad (1)$$

$$CR_i = \frac{FD_i}{\sum_j FD_j} \quad (2)$$

where i represents a POIs category, N_i is the count of type i POIs within a given parcel, and S_i is the total count of type i POIs in the entire dataset. CR_i is the proportion of the frequency density of type i relative to the sum of frequency densities for all POI types j within that parcel.

A parcel was designated as industrial if its Category Ratio for industrial POIs exceeded 0.5. These resulting sets of industrial parcels were then cross-referenced with the Xiaoshan District’s official Regulatory Detailed Plan and subjected to manual inspection. This final validation process yielded 77 industrial parcels, the spatial distribution of which is illustrated in Figure 3.

To construct the ground-truth inefficiency indicator, this study utilized data from the official 2022 and 2023 “Benefit-per-Mu”¹ Comprehensive Evaluation, a key policy instrument implemented by the Xiaoshan District government. The system classifies

enterprises into four classes: Class A (Priority Development), Class B (Encouraged for Improvement), Class C (Regulated for Rectification), and Class D (Slated for Elimination). This research targets the distribution of Class D enterprises, as they signify the lowest level of land use performance. A continuous inefficiency surface was subsequently generated by geocoding these Class D enterprises and applying a point density analysis. The final indicator for each parcel is defined as mean density value within its boundary, a method that provides statistical robustness against potential data omissions. The spatial distribution of this inefficiency indicator is illustrated in Figure 4.

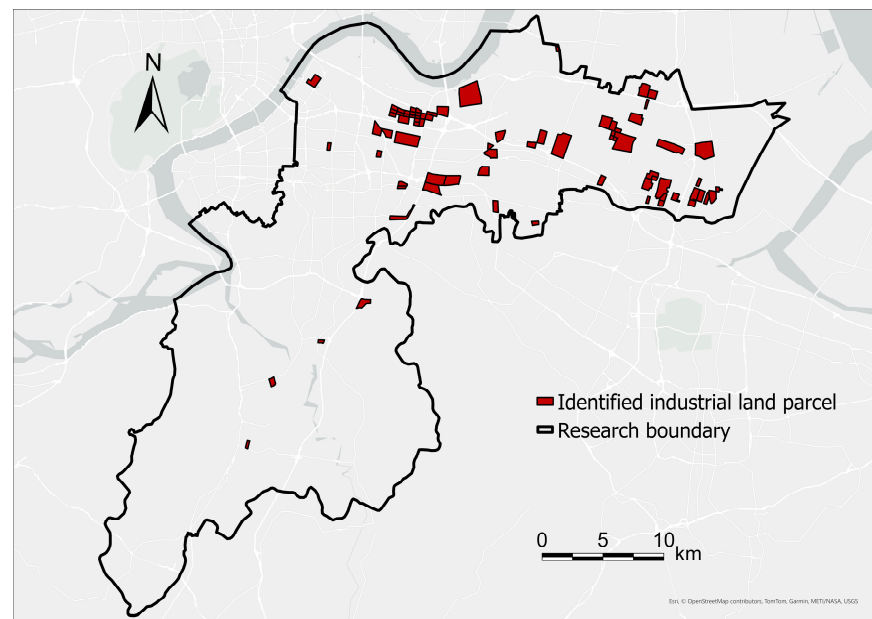


Figure 3. Spatial distribution of identified industrial land parcel.

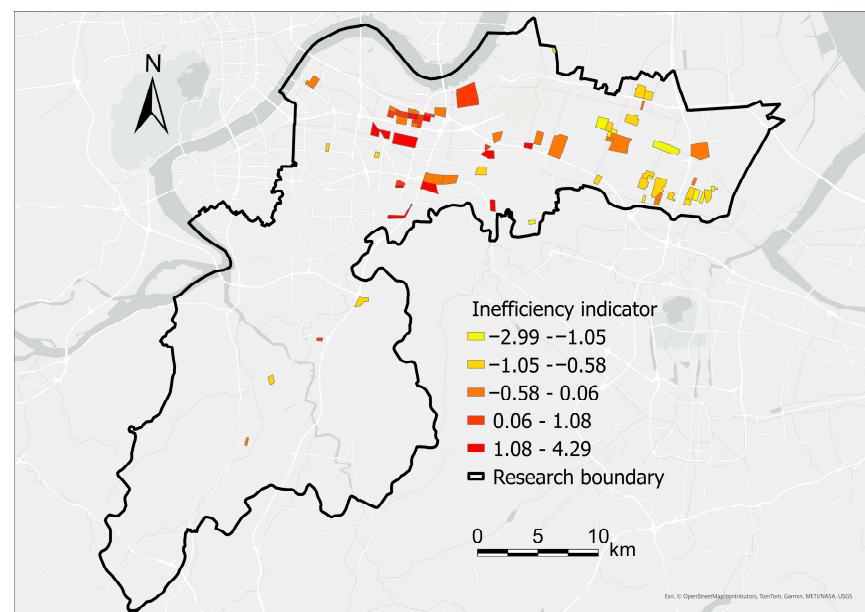


Figure 4. Spatial distribution of the inefficiency indicator of the industrial land parcel.

3.3. Preliminary Selection of the Evaluation System

To address the multidimensional requirements inherent in identifying inefficient industrial land and the systematic demands of performance evaluation, this study formulated

a problem-oriented evaluation system. Tailored to the specific characteristics of industrial land use in the research area, the evaluation system is structured around two primary dimensions: intrinsic parcel attributes and surrounding environmental attributes. The former focuses on indicators reflecting the inherent performance characteristics of the parcel itself, while the latter emphasizes the synergistic effects of the surrounding environment. The specific sub-indicators derived from these two dimensions are detailed in Table 1.

The selection of indicators, informed by a comprehensive review of the determinants of industrial land use efficiency, establishes a multi-dimensional evaluation system. The intrinsic attributes of a parcel are pivotal to its performance. Land use intensity, a critical factor, is quantified using the Floor Area Ratio (FAR), as lower intensity often signals underutilization and potential for redevelopment [35]. Economic value and potential are quantified by the Benchmark Land Price (BLP), as land prices directly reflect market perceptions of efficiency and are a central focus of land marketization reforms [36,37]. To quantify on-site economic vibrancy, Nighttime Light (NTL) intensity is employed as a reliable proxy for socio-economic activity [38]. Finally, the physical characteristics of the parcel are evaluated using the Landscape Shape Index (LSI), as parcel morphology and fragmentation can influence development feasibility and operational efficiency [18].

Beyond the parcel itself, surrounding environmental attributes exert a significant influence upon its efficiency. The location context is quantified by the Network Distance to Regional Centers (NDC), as accessibility and proximity to economic hubs are well-established determinants of land performance [39]. The quality of the local transport network is specifically measured by Road Betweenness Centrality (RBC), as efficient highway access is crucial for industrial operations [40]. The adequacy of supporting services is quantified by Infrastructure POI Kernel Density (IKD), given that infrastructure provision is a significant factor in industrial upgrading [41]. Lastly, the Residential & Industrial Land Use Mix (RIM) is incorporated to evaluate the degree of functional integration or separation, which represents a key consideration for optimal land allocation and sustainable urban form [42].

Table 1. Preliminary evaluation system for identifying inefficient industrial land.

Dimensions	Description	Indicators	Data Source
Intrinsic Parcel Attributes	Inadequate land use intensity	Floor Area Ratio (FAR) ↓	FAR of each parcel is calculated based on 3D-GloBFP (3D Global Building Footprints), which is a global dataset providing the height of individual buildings for the year 2020, created by integrating multi-source remote sensing data with advanced machine learning techniques [43].
	Low economic contribution and development potential	Benchmark Land Price (BLP) ↓	Hangzhou Municipal Bureau of Land and Resources.
	Low intensity of human and economic activity	Nighttime Light (NTL) ↓	2022 and 2023 annual average VIIRS nighttime lights data from the Earth Observation Group (EOG), a temporally consistent product with a 15 arc-second resolution derived from monthly cloud-free composites [44].
	High degree of parcel irregular shape fragmentation	Landscape Shape Index (LSI) ↑	Shape Index (SHAPE) metric calculated by Fragstats v4.2.1 software, which measures the regularity of a polygon's shape by averaging its edge curvature

Table 1. Cont.

Dimensions	Description	Indicators	Data Source
Surrounding Environmental Attributes	Inadequate planning and functional mix (work–life balance)	Residential & Industrial Land Use Mix (RIM) ↓	The Shannon entropy of residential and industrial POI from Amap, both for 2022 and 2023
	Remoteness and poor access to regional centers	Network Distance to Regional Centers (NDC) ↑	The network distance from each study parcel to regional centers (Qianjiang Century City and Xiaoshan Old Town), based on OpenStreetMap (OSM) road network data for 2022 and 2023
	Low traffic accessibility from inefficient road network layout	Road Betweenness Centrality (RBC) ↓	The betweenness centrality of each road node was calculated using Gephi 0.10.1 software based on the corrected OSM road networks for 2022 and 2023, and the average centrality value within each parcel was computed after spatial interpolation.
	Insufficient provision of regional infrastructure	Infrastructure POI Kernel Density (IKD) ↓	The kernel density of Amap POIs of related to urban infrastructure (e.g., public utilities, charging stations, gas stations, and other energy facilities), both for 2022 and 2023

Note: “↑” and “↓” refers to positive and negative correlation with inefficiency respectively.

3.4. Analytic Hierarchy Process and Random Forest

To construct the evaluation framework for industrial land performance and identify inefficient parcels, this study initially applied the Analytic Hierarchy Process (AHP) to establish a set of expert-derived weights across hierarchical indicators. These initial weights served as critical inputs for subsequent weight optimization procedures. AHP is a multi-criteria decision-making method that synthesizes qualitative judgment with quantitative analysis. The method involves decomposing the decision problem into a hierarchical structure, typically comprising a goal layer and one or more criteria layers.

This process entailed surveying urban planning and renewal experts with localized knowledge of the study area to construct pairwise comparison matrices ($A = [a_{ij}]$). Using a 1–9 fundamental scale, these matrices quantify the relative importance of each indicator in contributing to industrial land inefficiency with the reciprocal property $a_{ij} = 1/a_{ji}$. Subsequently, the normalized weights were derived from the principal eigenvector of each matrix, calculated using the Geometric Mean Method as follows:

$$\omega_i = \frac{\left(\prod_{j=1}^n a_{ij}\right)^{1/n}}{\sum_{k=1}^n \left(\prod_{j=1}^n a_{kj}\right)^{1/n}} \quad (3)$$

To ensure the reliability of expert judgments, the logical consistency of each matrix was validated. The Consistency Ratio (CR) was required to be less than 0.1. The CR depends on the principal eigenvalue ($\lambda_{\max}$), calculated as:

$$\lambda_{\max} = \frac{1}{n} \sum_{i=1}^n \frac{(A\omega)_i}{\omega_i} \quad (4)$$

Ultimately, the composite weights were synthesized by multiplying the local weights across the hierarchical levels. This methodology ensures mathematical consistency while

incorporating expert experience, providing a structured and transparent foundation for the subsequent machine learning-based weight optimization.

This study employs a Random Forest (RF) model to capture the complex relationships between parcel attributes and industrial land inefficiency indicator within the labeled parcel dataset. RF is selected because it can model complex interactions without requiring a predefined functional form, is robust to outliers and noise, and its inherent bootstrap aggregation reduces variance and helps reduce overfitting. Crucially, RF provides an internal out-of-bag (OOB) error estimate, which serves as an unbiased validation metric without splitting the dataset given the modest sample size. RF was implemented using the Scikit-learn library with optimized hyperparameters, including the number of trees ($n_estimators = 1000$) and maximum depth ($max_depth = 10$). For each segment, an RF is trained on the entire segment-specific dataset using bootstrapping without an explicit train–test split; generalization performance is assessed via the OOB Mean Absolute Error (MAE). This design maximizes data usage while providing a reliable check against overfitting.

3.5. Pattern-Sensitive Segmentation and Indicator Weight Optimization

This study introduces a novel, two-phase framework to calibrate subjective expert-derived weights with objective, data-driven insights. The first phase implements a pattern-sensitive segmentation to partition the dataset based on intrinsic data attributes. The second phase performs a segment-specific weight optimization, employing a Random Forest model to refine the initial AHP weights for each identified pattern. The complete process is outlined in Algorithm 1.

The first phase introduces a pattern-sensitive segmentation to address the spatial heterogeneity inherent in land performance data. The concept of ‘pattern-sensitive’ is central to this algorithm, moving beyond conventional methods that either assume spatial proximity dictates behavior (e.g., geographic clustering) or impose a single, uniform relationship across all parcels (e.g., global AHP). In this context, a ‘pattern’ is defined not by location, but as a distinct underlying relationship between the evaluation indicators (X) and the land performance outcome (y), which is considered a land inefficiency pattern in the study context. The segmentation is therefore ‘sensitive’ because it is designed to discover and isolate these latent functional relationships.

The proposed pattern-sensitive segmentation framework is designed to partition the entire dataset $D = (x_i, y_i)_{i=1}^N$ (where x_i is the multi-dimensional indicator vector for the plot i , and y_i is its performance) into K distinct segments, $C_1, \dots, C_K$. The determination of the optimal number of clusters, K , is a critical step. To address this, this study calculates the total Within-Cluster Sum of Squared Residuals across all clusters for a range of K values, to find the optimal value. By observing the plot of this metric against K , the minimum point of the U-shaped curve is selected as the optimal number of clusters. This point represents the optimal trade-off between segmentation effectiveness and model complexity. The underlying assumption is that within each segment C_k , the relationship between the indicators x and performance y is relatively consistent and linear, allowing it to be effectively modeled by a dedicated multiple linear regression model, $f_k(x) = w_k^T x + b_k$. The algorithm begins by utilizing K-Means for an initial partitioning of the feature space to obtain initial clusters. Subsequently, it enters an iterative optimization phase that alternately executes the following two core steps until convergence:

- A. **Model Fitting.** A dedicated linear regression model is fitted to the data points within each current cluster C_k . The model parameters (w_k, b_k) are adapted by minimizing the sum of squared residuals within the cluster:

$$(w_k, b_k) = \underset{w, b}{\operatorname{argmin}} \sum_{i: (x_i, y_i) \in C_k} \left(y_i - (w^T x_i + b) \right)^2 \quad (5)$$

- B. **Sample Reassignment.** For every data point (x_i, y_i) in the entire dataset, its prediction error with respect to each cluster's regression model $f_k(x)$ is calculated. The point is then reassigned to the cluster that yields the minimum error, updating its segment c_i :

$$c_i = \underset{k \in \{1, \dots, K\}}{\operatorname{argmin}} (y_i - f_k(x_i))^2 \quad (6)$$

This process is repeated iteratively until the cluster memberships stabilize, signifying that the algorithm has converged. In this manner, the algorithm automatically identifies and separates the latent, distinct response patterns within the data, grouping plots with similar land inefficiency patterns into the same category. This provides a refined and granular data foundation for the subsequent targeted optimization of indicator weights for each segment.

The second phase focuses on indicator weight optimization, in which a specialized trainable linear layer (referred to as the weight optimizer) is employed. This layer is designed to directly model the weighted-sum structure of the indicator framework, which is fundamentally a linear construct. The proposed weight optimizer has a total of 9 trainable parameters: 8 weight parameters ($W_{\text{optimizer}} = [w_i]_{1 \times 8}$), corresponding to the 8 indicators investigated in this study, and a bias parameter b . The output of the weight optimizer is calculated by the following equation:

$$\text{output} = \text{RELU}(XW^T - b) \quad (7)$$

where X is the input attribute vector, and W and b are the internal parameters of the optimizer. To preserve their role as proportional attributes, the weight parameters W are normalized within each training iteration prior to this output computation.

During initialization, the weight parameters W are set to expert-derived weights for the corresponding indicators, thereby embedding expert knowledge into the optimizer from the outset. Our objective is to minimize the discrepancy between the output distribution of the weight optimizer (which is more subjective but highly generalizable) and that of a Random Forest model (which is more objective and reflective of local characteristics). Furthermore, we constrain the weights within the optimizer to prevent them from deviating significantly from the initial expert-derived weights during training. Consequently, the loss function is formulated as follows:

$$\mathcal{L} = \left(1 - \text{PearsonCorr}(y_{\text{target}}, y_{\text{pred}}) \right) + \lambda \cdot \|w_{\text{optimizer}} - w_{\text{expert}}\|^2 \quad (8)$$

where $\mathcal{L}$ is the loss function of the weight optimizer; λ is a hyperparameter that controls the penalty for the divergence of the optimized weights from the expert-derived weights; $w_{\text{optimizer}}$ is the current weights in the weight optimizer after normalization; w_{expert} is the expert-derived weights (the initial weights); and y_{target} and y_{pred} are the outputs of the random forest and the weight optimizer on augmented input data, respectively.

For each segment k , the weight optimization phase of the algorithm is divided into two stages:

- A. **Training the Random Forest (RF) model.** The RF model is first trained on the labeled parcel data of the segment k ($\mathcal{D}_k$), where it captures the local data distribution from this limited dataset.
- B. **Training the weight optimizer.** To address limited labeled data, we employ a data augmentation strategy. Synthetic attribute vectors ($X_{\text{syn},k}$) are generated and merged

with the original labeled features to construct an augmented input matrix ($X_{aug,k}$). Corresponding target values (y_{target}) are subsequently obtained by feeding $X_{aug,k}$ into the pre-trained Random Forest (RF_k). A linear weight optimizer ($Optimizer_k$), initialized with expert-derived weights (W_{expert}), is then trained on $X_{aug,k}$ to fit y_{target} . This optimization minimizes a composite loss function ($\mathcal{L}$) comprising a Pearson correlation-based objective for target alignment and a regularization term to constrain deviations from expert-derived weights.

Algorithm 1 Framework for Segmented Weight Optimization

Procedure SegmentedWeightOptimization($\mathcal{D}_L, W_{expert}, \lambda, range_k$)

Input: Labeled dataset $\mathcal{D}_L = \{(X_L, y_L)\}$, expert-derived initial weights W_{expert} , regularization strength λ , range of cluster number $range_k$.

Output: A dictionary of trained weight optimizers $\mathcal{M}_{optimizers}$, partitioning models $\mathcal{M}_{partition}$.

Phase 1: Data Segmentation using Labeled Data

Initialize empty list total_SSE_list.

for k **in** $range_k$ **do**

 Initialize cluster labels via K-Means on attributes X_L .

repeat

 For each cluster, fit a Linear Regression model on its data subset from $\mathcal{D}_L$.

 Reassign each data point $(x_i, y_i) \in \mathcal{D}_L$ to the cluster whose model yields the minimum $(y_i - \hat{y}_i)^2$.

until cluster assignments converge

 Calculate the total Sum of Squared Errors (SSE) across all final clusters for the current k .

 Append SSE to total_SSE_list.

end for

Determine $k_{optimal}$ by applying the Elbow Method to the SSE curve based on total_SSE_list.

Perform the final Iterative Linear-Fit Clustering on $\mathcal{D}_L$ using $k_{optimal}$ to obtain final segment labels $labels_L$ and partitioning models $\mathcal{M}_{partition}$

Phase 2: Per-Segment Weight Optimization with Data Augmentation

Initialize empty dictionary $\mathcal{M}_{optimizers}$.

Initialize empty dictionary $\mathcal{W}_{optimizers}$.

for each unique segment ID k **in** $labels_L$ **do**

 Extract labeled data subset $\mathcal{D}_k = \{(X_k, y_k)\}$ from $\mathcal{D}_L$.

 Train a Random Forest on $\mathcal{D}_k$, noted as RF_k .

 Generate a set of synthetic attribute vectors $X_{syn,k}$.

 Create an augmented input $X_{aug,k} = X_k \cup X_{syn,k}$.

 Initialize a trainable linear layer with weights W_{expert} , noted as $Optimizer_k$.

 Define composite loss $\mathcal{L} = \left(1 - \text{PearsonCorr}(y_{target}, y_{pred})\right) + \lambda \cdot \left|W_{optimizer} - W_{expert}\right|_2^2$.

for a set number of training steps **do**

 Generate target values y_{target} by getting predictions from RF_k on input $X_{aug,k}$.

 Train $Optimizer_k$ on input $X_{aug,k}$ to match y_{target} by minimizing $\mathcal{L}$.

end for

Save the best version of the trained $Optimizer_k$ to $\mathcal{M}_{optimizers}[k]$.

Save the internal weights of the best – trained $Optimizer_k$ to $\mathcal{W}_{optimizers}[k]$.

end for

return $\mathcal{M}_{optimizers}$, $\mathcal{W}_{optimizers}$ and $\mathcal{M}_{partition}$

end procedure

4. Results

4.1. AHP Result

Based on the preliminary evaluation system, a questionnaire was constructed to collect expert knowledge and determine indicator weights using the AHP. To this end, four experts with extensive expertise in urban planning and redevelopment, as well as thorough familiarity with the study area, were solicited to complete the survey (detailed information on the survey questionnaire is presented in Appendix A).

Regarding the goal-layer indicators (dimensions), the AHP-derived weights were calculated as 72.16% for parcel attributes and 27.84% for surrounding environment attributes. The weights for the criteria-layer indicators associated with parcel attributes are detailed in Table 2. The resulting Consistency Ratio (CR) for this group was 0.051, which satisfies the consistency requirement. Furthermore, the weights for the criteria-layer indicators pertaining to surrounding environment attributes are shown in Table 3. The associated CR value was 0.016, likewise demonstrating acceptable consistency.

Table 2. Criteria layer indicators under parcel attributes.

Indicators	Problem Addressed by the Indicator	Weights
FAR	Inadequate land use intensity	25.00%
BLC	Low economic contribution and development potential	49.72%
NTL	Low intensity of human and economic activity	17.17%
LSI	High degree of parcel irregular shape fragmentation	8.11%

Table 3. Criteria layer for surrounding environment attributes.

Indicators	Problem Addressed by the Indicator	Weights
RIM	Inadequate planning and functional mix (work–life balance)	33.43%
NDC	Remoteness and poor access to regional centers	13.06%
RBC	Low traffic accessibility from inefficient road network layout	17.13%
IKD	Insufficient provision of regional infrastructure	36.38%

As shown in Table 4, the expert-derived weighting system exhibits a multi-dimensional structure with a distinct primary focus. Economic and spatial indicators occupy a dominant position, with BLP (35.88%) and FAR (18.0%) receiving the highest weights; this underscores the central importance placed on economic value and intensive land use in industrial renewal. Following these, social function indicators represented by NTL (12.39%), IKD (10.13%), and RIM (9.31%) were assigned moderate importance. Serving as crucial complements to the primary economic and spatial dimensions, these indicators augment the performance assessment by incorporating perspectives of economic vitality, public service provision, and the job–housing balance. In contrast, indicators related to transport accessibility and parcel morphology received consistently lower weights. This suggests that experts likely perceive issues such as poor accessibility or parcel fragmentation as symptoms rather than causes of land use inefficiency. Furthermore, the perceived influence of these factors is likely sensitive to the specific conditions of the study area and the scale of analysis.

4.2. Pattern-Sensitive Segmentation

The proposed algorithm was applied to optimize expert-derived indicator weights. The first phase involved partitioning the dataset based on land use inefficiency patterns. To determine the optimal number of segments (K), the algorithm's performance was evaluated for K-values ranging from 2 to 11, as shown in Table 5. The Sum of Squared Errors (SSE) was minimized when the industrial parcels were divided into 4 classes (SSE = 4.46).

Consequently, the dataset was partitioned into four segments. Additionally, a weight optimization without the segmentation process (denoted as global training) was conducted to comparatively demonstrate the necessity of data segmentation.

Table 4. Expert-derived weights of the preliminary evaluation system.

Indicators	Problem Addressed by the Indicator	Weights	Ranks
FAR	Inadequate land use intensity	18.04%	2
BLP	Low economic contribution and development potential	35.88%	1
NTL	Low intensity of human and economic activity	12.39%	3
LSI	High degree of parcel irregular shape fragmentation	5.85%	6
RIM	Inadequate planning and functional mix (work–life balance)	9.31%	5
NDC	Remoteness and poor access to regional centers	3.64%	8
RBC	Low traffic accessibility from inefficient road network layout	4.77%	7
IKD	Insufficient provision of regional infrastructure	10.13%	4

Table 5. Sum of Squared Errors (SSE) for different K-values.

K	SSE
2	12.434123
3	6.9906654
4	4.458191
5	15.245002

Figure 5 illustrates the spatial distribution of these four segments. Segments S1 and S2 are predominantly distributed in the central-eastern part of the study area, exhibiting a relatively dispersed and overlapping distribution. Conversely, S4 is primarily situated in the central region and S3 is mainly concentrated in the western region, with both displaying a more concentrated spatial pattern.

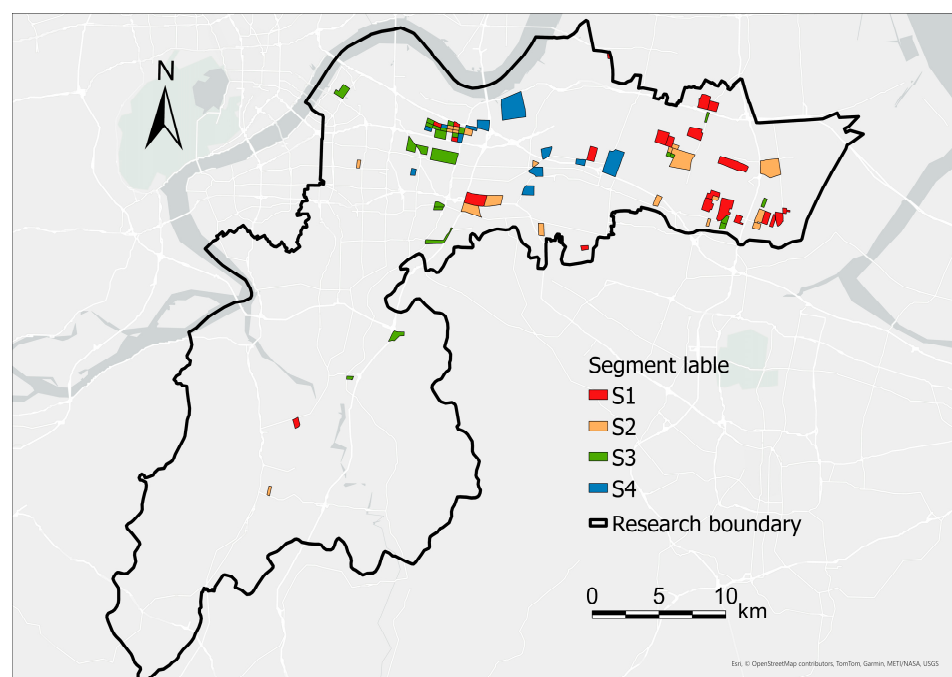


Figure 5. Segments of the industrial land parcel.

4.3. Indicator Weight Optimization

The second phase of the algorithm refines the weights within each segment using a Random Forest model to introduce data-driven objectivity. To evaluate the training result of the Random Forest model, their OOB performance is evaluated: training Pearson correlations ranged from 0.946 to 0.981, while OOB MAE (ranging from 0.33 to 1.05) remained relatively modest, confirming the reliability of the training results. A penalty term constrains these adjustments, ensuring that the final weights remain aligned with the original expert knowledge.

Following the training of the weight optimizers, an inefficiency score was calculated for each parcel by applying a weighted sum to the standardized attributes using the optimized weights ($I = WTX$, I is the inefficiency score, W is the optimized weight, and X is the parcel indicator vector). To validate the performance of the optimized system, we assessed the correlation between these calculated scores and the observed inefficiency indicators. Across the four segments, the Pearson correlation coefficients ranged from 0.66 to 0.82, Spearman's coefficients ranged from 0.44 to 0.72, and Kendall's coefficients ranged from 0.33 to 0.56. All correlations were statistically significant ($p < 0.05$). These results demonstrate that the evaluation systems, refined by our algorithm, effectively capture the distribution patterns of actual land use inefficiency.

We benchmarked our model against four alternatives: AHP (expert-derived weights), EWM-AHP (AHP combined with EWM), HBN-AHP (AHP combined with HBN model), and the global training approach (without segmentation). The results revealed critical limitations in these conventional methods (Table 6). In segments S1, S2, and S3, the three AHP-based benchmarks yielded scores that were significantly and negatively correlated with the observed inefficiency data. In segment S4, the AHP method produced scores with a relatively high positive correlation to the observed data (Pearson's $r = 0.51$). The modification of EWM-AHP further strengthened the positive correlation with statistical significance (Pearson's $r = 0.70$, $p < 0.05$), while HBN-AHP substantially reduced the positive correlation (Pearson's $r = 0.01$). These findings demonstrate that both pure expert judgment and its conventional modifications exhibited significant limitations in adequately reflecting the actual data distribution within our study area. Furthermore, the results underscored the necessity of data segmentation, that the global training model exhibited limitations similar to the other three benchmarks, reflecting a significant negative correlation in S1, S2 and S3, while showing an insignificant positive correlation in S4. The global training model failed to capture distinct correlation patterns, thus validating the necessity of the data segmentation process in our algorithm.

4.4. Comparative Analysis of Weight Optimization Results

The optimized indicator weights reveal significant spatial heterogeneity in the drivers of land use inefficiency, contrasting with the uniform structure of the global AHP framework (Table 7). Notably, the dominant role of BLP is diminished. While the AHP assigned BLP a weight of 35.88%, our algorithm adjusts it to a range of 18.21–27.31%. This suggests that expert-derived weight may overshadow other critical drivers in certain contexts. Furthermore, the algorithm elevates indicators considered secondary by experts, such as IKD, RIM and LSI, to primary roles in specific segments, confirming that the drivers of inefficiency vary spatially.

Specifically, in S1, the algorithm produces a balanced model in which IKD (20.39%), RIM (20.40%), RBC (19.18%), LSI (18.90%) and BLP (18.21%) receive similarly high weights. This contrasts with the AHP model's primary focus on BLP. A key counter-intuitive finding is the strong negative weight for NDC (−15.44%) in S1 (and similarly in S2, albeit with a weaker weight around −3.46%). This suggests that within these segments,

proximity to urban centers correlates with higher inefficiency. This phenomenon is likely attributable to land speculation: parcels near urban cores (such as Qianjiang Century City) possess higher redevelopment potential, and consequently landowners anticipate lucrative rezoning for commercial or residential use, which disincentivizes long-term industrial investment and leads to operational stagnation of industrial lands, thereby resulting in land use inefficiency.

Table 6. The performance of our algorithm against benchmarks.

Segment	Model	Pearson_corr	Pearson_p	Spearman_corr	Spearman_p	Kendall_corr	Kendall_p
S1	AHP	−0.83	0.0000	−0.56	0.0064	−0.39	0.0118
	EWM-AHP	−0.84	0.0000	−0.56	0.0070	−0.38	0.0140
	HBN-AHP	−0.73	0.0001	−0.57	0.0059	−0.42	0.0058
	Global training	−0.48	0.0241	−0.49	0.0221	−0.35	0.0228
	Our algorithm	0.71	0.0002	0.44	0.0394	0.33	0.0309
S2	AHP	−0.68	0.0011	−0.64	0.0026	−0.47	0.0030
	EWM-AHP	−0.69	0.0008	−0.65	0.0021	−0.49	0.0018
	HBN-AHP	−0.63	0.0030	−0.56	0.0103	−0.36	0.0164
	Global training	−0.58	0.0068	−0.60	0.0053	−0.41	0.0111
	Our algorithm	0.82	0.0000	0.72	0.0003	0.56	0.0004
S3	AHP	−0.64	0.0009	−0.54	0.0074	−0.34	0.0219
	EWM-AHP	−0.65	0.0008	−0.57	0.0048	−0.35	0.0189
	HBN-AHP	−0.58	0.0040	−0.63	0.0012	−0.43	0.0043
	Global training	−0.88	0.0000	−0.82	0.0000	−0.59	0.0000
	Our algorithm	0.66	0.0006	0.60	0.0025	0.45	0.0026
S4	AHP	0.51	0.0880	0.54	0.0709	0.33	0.1526
	EWM-AHP	0.70	0.0113	0.70	0.0114	0.55	0.0138
	HBN-AHP	0.01	0.9745	0.05	0.8854	0.06	0.8168
	Global training	0.39	0.2063	0.33	0.2969	0.27	0.2496
	Our algorithm	0.72	0.0082	0.70	0.0106	0.53	0.0161

Table 7. The optimized indicator weights of our algorithm and AHP.

Segment	Model	FAR	RBC	BLP	NTL	LSI	RIM	IKD	NDC
S1	Our algorithm	9.42%	19.18%	18.21%	8.95%	18.89%	20.40%	20.39%	−15.44%
S2		13.76%	−2.90%	20.99%	7.45%	11.28%	21.84%	31.03%	−3.46%
S3		5.61%	−9.19%	27.31%	17.15%	22.05%	24.88%	5.43%	6.76%
S4		18.10%	11.22%	23.42%	1.88%	15.50%	13.55%	13.06%	3.27%
All segments	AHP	18.04%	4.77%	35.88%	12.39%	5.85%	9.31%	10.13%	3.64%

In S2, social and service-related factors are dominant. IKD holds the highest weight (31.03%), followed by RIM (21.84%) and BLP (20.99%). This possibly indicates that inefficiency in this region is primarily driven by severe “industry–city separation”, a mismatch between industrial activities and supporting urban infrastructure. These parcels are possibly located in production-oriented zones, reflecting an outdated planning model that prioritized industry over livability, ultimately limiting both the enterprise performance and the overall value of the land.

In S3, the model highlights a combination of economic value (BLP, 27.31%), functional mix (RIM, 24.88%), and physical form (LSI, 22.05%). The significant negative weight for RBC (−9.19%) suggests that in older industrial zones, high network centrality may be a detrimental factor rather than an asset. Such locations often become hotspots for land-use competition, attracting commercial and residential development. This pressure can inflate land values and create an uncertain planning environment, discouraging long-term investment in industrial upgrades and leading to operational stagnation and functional inefficiency.

The S4 weights most closely resemble the AHP model's emphasis on economic and physical attributes, with BLP (23.42%) and FAR (18.10%) identified as the top indicators. This might suggest that inefficiency in this segment stems from more conventional issues like enterprise underperformance itself or insufficient land use intensity. The high weight assigned to FAR and BLP provides empirical support for local policies advocating vertical industrial development, which are designed to intensify industrial land use to promote efficiency and reduce land costs within industrial parks. Notably, a critical divergence between the expert-derived weights and the algorithm result is the negligible weight of NTL (1.88%), which experts ranked third (12.39%). This suggests that in S4, which likely contains modern industrial parks, NTL is a poor proxy for economic activity due to automated processes or single-shift operations. The algorithm's ability to identify this context-specific relationship is a key advantage.

In summary, the comparative analysis of indicators across segments highlights the algorithm's utility. It transforms static, universally applied expert weights into dynamic, context-aware parameters. The emergence of negative weights and the significant reordering of indicator importance across segments provide strong evidence that a segmented, data-driven approach is essential for accurately diagnosing the spatially heterogeneous nature of inefficient industrial land use.

Additionally, the analysis identifies four possible distinct profiles of industrial land inefficiency. Inefficiency in S1 appears to be driven by land speculative holding in valuable locations, while in S3 it possibly stems from land use competition in accessible areas. S2 shows that inefficiency is likely caused by inadequate urban services, resulting from possible industry–city separation. Finally, S4 might represent inefficiency arising from insufficient land use intensity, supporting the need for local policies that promote vertical industrial development.

4.5. Spatial Distribution of Industrial Inefficiency Score

To examine differences in spatial distribution patterns between the inefficiency scores derived from expert-based AHP weights and those from optimized weights, we mapped their normalized distributions based on Z-score. The AHP-derived inefficiency scores (Figure 6) exhibited a clear spatial gradient, decreasing from east to west, with the majority of highly inefficient industrial land parcels concentrated in zones S1 and S2 (the eastern part). In contrast, the scores derived from the optimized weights (Figure 7) were more effective in identifying inefficient parcels in zone S3 (central-western part of the area) while indicating lower inefficiency in the eastern region (mainly S1 and S2). This distribution corresponds more closely to the pattern of the actual inefficiency indicator (Figure 4).

Figure 8 presents boxplots comparing the distributions of the inefficiency indicator, the expert-based AHP score, and our algorithm's optimized score across four segments, with all values standardized using z-scores. The analysis reveals significant discrepancies for the expert-based AHP method. Overall, the ground truth inefficiency indicator exhibited the highest median in segment S3 (≈ 0.7) and the lowest in S1 (≈ -0.8). In stark contrast, the inefficiency scores derived from expert-based weights showed an opposing trend, with the highest median occurring in S1 (≈ 1.0) and the lowest in S3 (≈ -0.8), indicating a significant discrepancy with the actual data distribution. However, the inefficiency scores from the optimized weights aligned more closely with the ground truth. They correctly identified S3 as the segment with the highest median score (≈ 0.2) and S1 with the lowest (≈ -0.2), although the variation in these medians across segments was narrower compared to the actual inefficiency indicator.

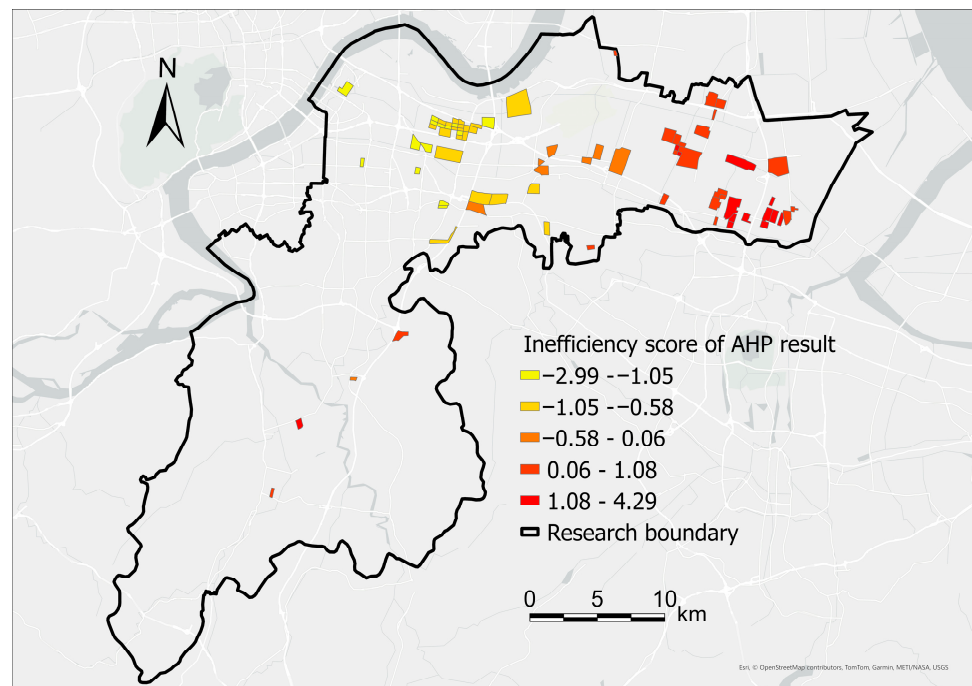


Figure 6. Spatial distribution of inefficiency score from AHP.

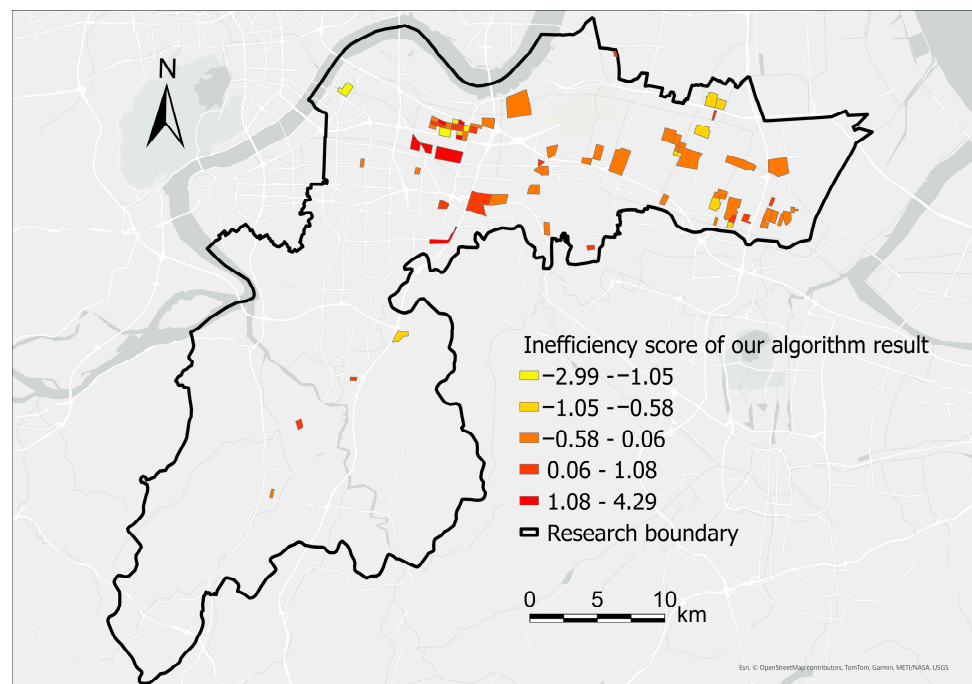


Figure 7. Spatial distribution of inefficiency score from our algorithm.

Specifically, in segment S1, where the actual inefficiency indicator is low (median ≈ -0.8), the AHP inefficiency score is paradoxically high (median ≈ 1.0). Our algorithm bridges this gap, producing a more aligned median inefficiency score (≈ -0.2). For S2, the median inefficiency score of AHP (≈ 0.4) significantly overestimated the median of the actual inefficiency indicator (≈ 0.0). In contrast, the inefficiency score from the optimized weights (≈ 0.0) demonstrated better alignment with the ground truth. Similarly, for S3, the expert weights produced a median inefficiency score (≈ -0.8) that substantially underestimated the ground truth median (≈ 0.7); conversely, the optimized weights yielded a much closer median inefficiency score (≈ 0.2) and exhibited a comparable interquartile range (IQR), indicating a more accurate

representation. Likewise, for segment S4, the inefficiency score of AHP (median ≈ -0.7) again underestimated the actual value (median ≈ 0.3), whereas the optimized inefficiency score was more proximate (median ≈ -0.1) and showed reduced variability, as evidenced by a small IQR. These results demonstrate that the weight optimization significantly enhances the model's capacity to reflect the true spatial patterns of inefficiency.

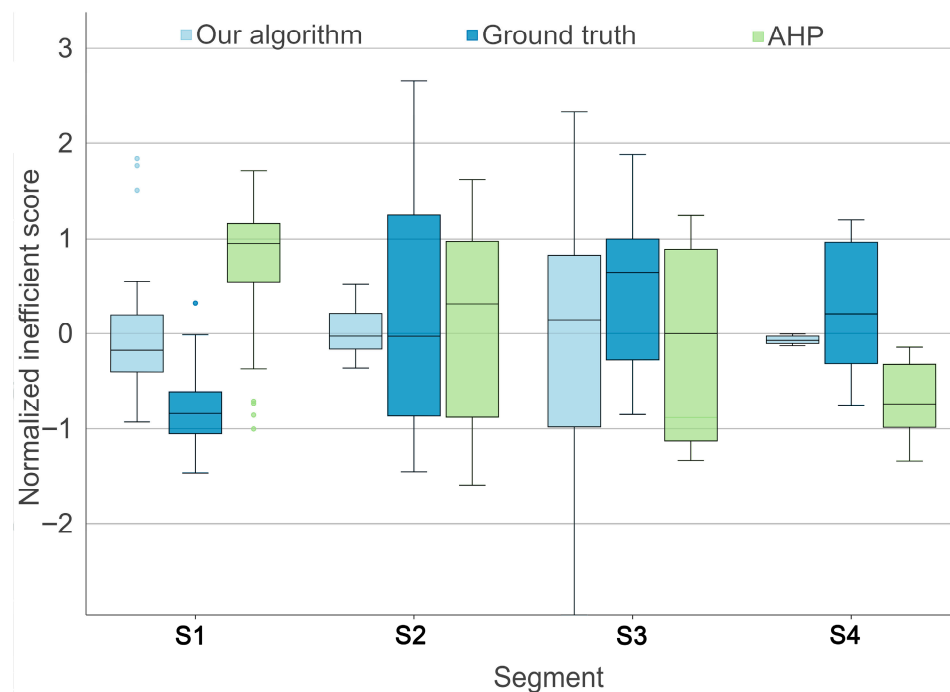


Figure 8. The inefficient score of the result of AHP, our algorithm and ground truth, grouped by segment.

4.6. Verifying the Temporal Generalization of the Optimized Weights

To validate the temporal generalization of the optimized indicator weights, we constructed a dataset for the year 2023. Subsequently, we calculated the inefficiency scores using our optimized weights and all benchmarks, assessing their correlation with the 2023 ground truth inefficiency indicator; these results are presented in Table 8. As indicated, our optimized weights sustained a significant positive correlation with the ground truth across all segments (Pearson's r ranging from 0.58 to 0.66). In contrast, the expert-based (AHP) weights and EWM-AHP methods persisted in exhibiting significant negative correlations in segments S1, S2, and S3, along with a positive but non-significant correlation in S4. Notably, the HBN-AHP model demonstrated inconsistent performance on the 2023 dataset: the negative correlations in S1 and S2 weakened to non-significant levels (Pearson's r from -0.13 to -0.08), while S3 and S4 exhibited non-significant positive correlations (Pearson's r from 0.17 to 0.47). Overall, the benchmark's capacity to reflect the ground truth remained limited. Consequently, the predictive performance of our optimized weights regarding the 2023 inefficiency indicator was substantially superior to that of the benchmarks, suggesting that the optimized evaluation systems possess strong temporal generalization capabilities within the study area.

Table 8. The performance of our algorithm against benchmarks on 2023 dataset.

Segment	Model	Pearson_corr	Pearson_p	Spearman_corr	Spearman_p	Kendall_corr	Kendall_p
S1	AHP	−0.83	0.0000	−0.58	0.0050	−0.38	0.0140
	EWM-AHP	−0.83	0.0000	−0.60	0.0031	−0.41	0.0069
	HBN-AHP	−0.13	0.5786	−0.22	0.3172	−0.19	0.3056
	Our Algorithm	0.66	0.0008	0.31	0.1652	0.23	0.1439
S2	AHP	−0.66	0.0015	−0.59	0.0058	−0.46	0.0038
	EWM-AHP	−0.67	0.0012	−0.60	0.0049	−0.45	0.0047
	HBN-AHP	−0.08	0.7294	−0.12	0.6274	−0.10	0.6143
	Our Algorithm	0.64	0.0023	0.39	0.0923	0.24	0.1458
S3	AHP	−0.62	0.0017	−0.53	0.010	−0.32	0.0335
	EWM-AHP	−0.63	0.0014	−0.51	0.0123	−0.33	0.0292
	HBN-AHP	0.17	0.4408	0.10	0.6616	0.08	0.6511
	Our Algorithm	0.59	0.0032	0.54	0.0083	0.36	0.0162
S4	AHP	0.50	0.0975	0.50	0.0952	0.33	0.1526
	EWM-AHP	0.45	0.1387	0.42	0.1745	0.30	0.1969
	HBN-AHP	0.47	0.1240	0.46	0.1334	0.40	0.1013
	Our Algorithm	0.58	0.0492	0.47	0.1108	0.30	0.1969

5. Discussion

This study introduced and validated a novel, pattern-sensitive framework to construct a set of parcel-scale evaluation systems for evaluating inefficient industrial land. The results demonstrate that our segmented, machine learning-optimized approach significantly outperforms conventional methods that rely on spatially uniform evaluation criteria. A key finding was the severe underperformance of the traditional AHP model and its common hybrid variants; their evaluation scores were negatively correlated with ground-truth inefficiency in three of the four identified segments. This provides strong empirical evidence against the “one-size-fits-all” evaluation paradigm. By overcoming this limitation, our framework successfully identified four distinct inefficiency patterns, demonstrating that the drivers of underperformance are highly heterogeneous across different local contexts.

Our findings empirically challenge the assumption of spatial homogeneity that underpins established evaluation methods like AHP [19,20] as well as its variants. While previous studies have noted the theoretical limitations of applying global weights in complex urban systems [21], our work provides a clear quantification of these limitations. Although other research has attempted to improve on the subjectivity of AHP by integrating it with objective weighting schemes like the Entropy Weight Method [33], Bayesian networks [34], or by using TOPSIS [23]. However, our comparative benchmark analysis demonstrates that these hybrid models exhibited unsatisfactory performance in our study area. Their fundamental limitation stems from their reliance on a single set of global weights and they are therefore unable to address the core problem of pattern heterogeneity. This underscores the first major advantage of our approach: the pattern-sensitive segmentation. By partitioning parcels based on shared functional relationships between indicators and outcomes rather than simple geography. Our algorithm directly models this heterogeneity, creating homogenous subgroups where evaluation is more reliable.

The second major advantage lies in the synergy between expert knowledge and machine learning to achieve data-driven objectivity. Unlike methods such as EWM, which assign weights based on formulaic, context-blind metrics like statistical dispersion [22,23], our algorithm uses a Random Forest model to capture the objective, often non-linear relationships directly from the local data distribution within each pattern. These data-driven insights are then used to systematically calibrate the initial, subjective expert-derived weights. This process grounds expert intuition within empirical, localized evidence,

overcoming the limitations of both purely subjective models (AHP) and formulaically objective ones (EWM). The ability to generate negative weights (e.g., for NDC in S1 and RBC in S3) and to significantly reorder indicator importance across patterns highlights the power of this data-driven calibration.

The third advantage of the pattern-sensitive framework is its capacity to identify the distinct drivers of inefficiency for each identified pattern to validate diagnostic capacity of the method. In segment S1, for instance, the framework identifies proximity to regional centers (NDC) as a primary driver of inefficiency. This finding contradicts the conventional view that accessibility enhances efficiency [39] and instead suggests a mechanism of land speculation [14]: proximity to urban cores and high redevelopment potential incentivize landowners to hold land in anticipation of lucrative rezoning for commercial or residential use, leading to disinvestment and operational stagnation of industrial activities, advocating for targeted policies such as anti-speculative taxation rather than ineffective infrastructure upgrades. The diagnostic insights for segment S2 (“industry–city separation”) are consistent with studies that link monofunctional industrial planning with poor land performance due to insufficient residential and service amenities whereas industry–city integration improves land use efficiency through functional complement and reduced spatial mismatch, indicating strategies such as targeted infrastructure investment [42]. The “land-use competition” diagnosed in segment S3 also aligns with studies indicating that land use competition intensifies in older industrial zones where high network centrality attracts commercial and residential development, inflating land values and creating planning uncertainty and thereby discouraging long-term industrial investment and leads to operational stagnation and functional inefficiency, suggesting land-use strategies such as stronger industrial zoning protections [45]. The framework also confirms established findings. For pattern S4, it identifies low Floor Area Ratio (FAR) as the key determinant, aligning with previous research on insufficient land use intensity [12,46] and empirically validating policies that promote vertical industrial development for this specific pattern. This capacity to generate specific, data-driven diagnoses for each pattern transforms the evaluation from a simple scoring exercise into a diagnostic process, enabling a shift from generalized policies to precise, evidence-based interventions. While these interpretations are derived from the algorithm’s output rather than pre-registered hypotheses, they are grounded in established economic and planning mechanisms and are presented as context-specific diagnostic suggestions.

However, several limitations of this study should be acknowledged, which in turn suggest directions for future research. First, the reliance on open-source data limits the precision of industrial parcel identification and characterization. The delineation of industrial land using OSM road networks and POI data lacks the geometric accuracy required for granular, enterprise-scale analysis. Furthermore, indirect indicators such as Nighttime Light imagery proved insufficient for capturing economic performance in S4 (highly automated industrial zones). Future work could integrate more direct, firm-level data on economic output, tax contributions, or energy consumption to refine the evaluation system. Second, our “ground truth” indicator was based on the density of government-designated “Class D” enterprises. While this is a robust and policy-relevant measure, it is specific to the local context. Future studies could test the framework using alternative performance metrics, such as environmental pollution levels or employment density. Third, while our algorithm identifies and models distinct data patterns, the interpretation of these patterns as specific archetypes (e.g., “land speculation”) relies on inference and would benefit from qualitative field verification. Fourth, while the modest sample size (77 parcels) reflects a case-specific constraint from using open-source data, our study demonstrates adequate framework performance under small-sample conditions due to the low-dimensional in-

indicator space; future studies may adopt more detailed datasets to further validate its stability and generalizability as a larger sample size will potentially improve the model performance. Finally, while the framework is designed to be generalizable, the specific archetypes identified are likely unique to Xiaoshan's development context. Applying this framework to other cities with different economic structures, such as resource-based cities or service-dominated economies, would be an important next step to test its robustness and identify a broader typology of industrial land inefficiency.

6. Conclusions

This study addresses a critical limitation in industrial land inefficiency evaluation: the reliance on spatially uniform, “one-size-fits-all” evaluation systems, and conventional weighting methods are either relatively subjective, as in the Analytic Hierarchy Process (AHP), or are based on mechanistic objectivity that ignores the ground truth data distributions. Our findings empirically demonstrate that AHP and its common hybrid variants are often misaligned with ground-truth inefficiency data.

To address these challenges, this study proposes and validates a novel framework that addresses these shortcomings through three main contributions. Primarily, we develop a replicable, open-source framework leveraging geospatial big data for parcel-scale analysis, providing the granularity essential for targeted urban renewal. Second, the framework introduces a novel segmentation method that identifies distinct inefficiency patterns by grouping parcels based on latent functional relationships between indicators and outcomes, replacing the flawed assumption of spatial homogeneity. Third, a hybrid weight optimization process is presented, where machine learning systematically calibrates initial expert-derived weights against objective, local data distributions within each identified pattern. The framework's effectiveness is demonstrated through a case study in Xiaoshan, which yields several context-dependent inefficiency patterns as illustrative outputs.

By enabling a shift from generalized policies to precise, evidence-based interventions, the pattern-sensitive framework serves as a robust and replicable tool for guiding sustainable urban renewal. While acknowledging the need for further validation in diverse urban contexts, this research establishes a new paradigm for industrial land inefficiency evaluation by effectively synthesizing domain expertise with data-driven objectivity through pattern-sensitive segmentation.

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Conflicts of Interest: Author Mingyu Zhang was employed by the company The Architectural Design & Research Institute of Zhejiang University Co., Ltd. The remaining authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Abbreviations

The following abbreviations are used in this manuscript:

3D-GloBFP	3D Global Building Footprints
AHP	Analytic Hierarchy Process
BLP	Benchmark Land Price
CR	Consistency Ratio
DEA	Data Envelopment Analysis
EOG	Earth Observation Group
EWM	Entropy Weight Method
FAR	Floor Area Ratio
FD	Frequency Density
IKD	Infrastructure POI Kernel Density
LSI	Landscape Shape Index
MCDA	Multi-Criteria Decision Analysis
MCDM	Multi-Criteria Decision-Making
NDC	Network Distance to Regional Centers
NTL	Nighttime Light
OSM	OpenStreetMap
POI	Points of Interest
RBC	Road Betweenness Centrality
RF	Random Forest
RIM	Residential & Industrial Land Use Mix
SFA	Stochastic Frontier Analysis
SHAPE	Shape Index
SSE	Sum of Squared Errors
TOPSIS	Technique for Order Preference by Similarity to Ideal Solution
VIIRS	Visible Infrared Imaging Radiometer Suite

Appendix A. Expert Consultation Questionnaire

Dear Expert,

We succinctly invite you to participate in this expert consultation survey regarding the evaluation of industrial land use efficiency.

Background & Purpose: In response to the strategic shift of urban development towards stock regeneration, the identification and redevelopment of underutilized industrial land have become critical priorities for sustainable urban renewal. This study aims to construct a robust, multi-dimensional evaluation framework to accurately identify inefficient industrial land at the parcel level. We are employing the Analytic Hierarchy Process (AHP) to determine the relative weights of evaluation indicators, and your professional judgment is invaluable in establishing a scientific basis for this system.

Research Area Profile: The empirical focus of this research is Xiaoshan District, Hangzhou. As a prominent manufacturing hub in the Yangtze River Delta, Xiaoshan is navigating the dual pressures of industrial upgrading and urban expansion. The district features a heterogeneous industrial landscape, ranging from traditional manufacturing clusters to modern high-tech parks, making the accurate diagnosis of land use inefficiency a complex but essential task.

Instructions: Based on your professional experience and understanding of urban planning, please compare the relative importance of the indicators in each pair listed below. Mark the option that best represents your judgment. All responses will be used solely for academic research purposes and will remain strictly confidential. Thank you for your time and contribution.

Rating Scale:

1. Significantly More Important: The left indicator is critical compared to the right.
2. Slightly More Important: The left indicator is moderately favored.
3. Equally Important: Both indicators contribute equally.
4. Slightly Less Important: The right indicator is moderately favored.
5. Significantly Less Important: The right indicator is critical compared to the left.

Part 1: Primary Dimensions Comparison

Q1. Comparative Importance of Main Dimensions: Please compare the importance of Intrinsic Parcel Attributes versus Surrounding Environmental Attributes in the context of identifying inefficient industrial land.

Dimension A (Intrinsic Parcel Attributes): Factors inherent to the parcel itself (e.g., intensity, economic output, shape).

Dimension B (Surrounding Environmental Attributes): Factors related to the parcel's external context (e.g., location, transport, facilities).

X	Significantly More Important	Slightly More Important	Equally Important	Slightly Less Important	Significantly Less Important	Y
Dimension A (Intrinsic Parcel Attributes)						Dimension B (Surrounding Environmental Attributes)

Part 2: Comparison of Intrinsic Parcel Attributes (Dimension A)

Q2. Comparative Importance of Sub-indicators under Intrinsic Parcel Attributes: Please compare the following indicators relative to each other within Dimension A.

Indicator A1—Land Use Intensity: Reflects the density of development (e.g., Floor Area Ratio/FAR).

Indicator A2—Economic Contribution & Potential: Measures the economic value and market potential of the land (e.g., Benchmark Land Price).

Indicator A3—Socio-economic Activity: Reflects the intensity of human and production activity (e.g., Nighttime Light Intensity).

Indicator A4—Parcel Fragmentation: Measures the irregularity of the parcel's shape (e.g., Landscape Shape Index).

X	Significantly More Important	Slightly More Important	Equally Important	Slightly Less Important	Significantly Less Important	Y
Indicator A2						Indicator A1
Indicator A3						Indicator A1
Indicator A3						Indicator A2
Indicator A4						Indicator A1
Indicator A4						Indicator A2
Indicator A4						Indicator A3

Part 3: Comparison of Surrounding Environmental Attributes (Dimension B)

Q3. Comparative Importance of Sub-indicators under "Surrounding Environmental Attributes": Please compare the following indicators relative to each other within Dimension B.

Indicator B1—Functional Mix (Work-Life Balance): Reflects the balance of residential and industrial functions and planning completeness.

Indicator B2—Proximity to Centers: Distance to regional hubs (e.g., Xiaoshan Old Town, Qianjiang Century City).

Indicator B3—Traffic Accessibility: Reflects the rationality of the transport layout and road network centrality.

Indicator B4—Infrastructure Provision: Measures the density of supporting facilities (e.g., public utilities, energy facilities).

X	Significantly More Important	Slightly More Important	Equally Important	Slightly Less Important	Significantly Less Important	Y
Indicator B2						Indicator B1
Indicator B3						Indicator B1
Indicator B3						Indicator B2
Indicator B4						Indicator B1
Indicator B4						Indicator B2
Indicator B4						Indicator B3

Note

- ¹ The **mu** is a Chinese unit of area, equivalent to approximately 666.67 square meters (m²) or 1/15 of a hectare (ha). The term “Benefit-per-Mu” is a direct translation of the official Chinese policy name “**mǔ chǎn xiào yì**”, a key metric that evaluates economic output per unit of land.

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